$$C^{(21)\,\mathrm{i1\_i2\_}}_{d\phi_2} \to$$
$$\frac{1}{16\pi^2}\Big(-\frac{1}{4} m_1 \tilde\mu g_1^4 y_d^{i1i2} \mathrm{LF}_{2,2,0}[m_1, \tilde\mu] - \frac{1}{12} m_2 \tilde\mu g_2^4 (9 y_d^{i1i2} + 4 y_d^{pi2} \delta_{pi1}) \mathrm{LF}_{2,2,0}[m_2, \tilde\mu] +$$
$$\frac{1}{6} m_2 \tilde\mu g_2^4 y_d^{pi2} \mathrm{LF}_{3,2,-1}[m_2, \tilde\mu] \delta_{pi1} + \frac{1}{18} \tilde\mu g_1^2 y_d^{pr} \overline{a_d}^{pr} (-3 y_d^{i1i2} + y_d^{si2} \delta_{si1} + 2 y_d^{i1s} \delta_{si2})$$
$$\mathrm{LF}_{2,2,0}[m_{\tilde d}^{r}, m_{\tilde q}^{p}] + \frac{1}{36} \tilde\mu g_1^2 y_d^{pr} \overline{a_d}^{pr} (3 y_d^{i1i2} - y_d^{si2} \delta_{si1} - 2 y_d^{i1s} \delta_{si2}) \mathrm{LF}_{3,2,-1}[m_{\tilde d}^{r}, m_{\tilde q}^{p}] +$$
$$\frac{1}{18} \tilde\mu g_1^2 y_e^{pr} \overline{a_e}^{pr} (-3 y_d^{i1i2} + y_d^{si2} \delta_{si1} + 2 y_d^{i1s} \delta_{si2}) \mathrm{LF}_{2,2,0}[m_{\tilde e}^{r}, m_{\tilde l}^{p}] +$$
$$\frac{1}{36} \tilde\mu g_1^2 y_e^{pr} \overline{a_e}^{pr} (3 y_d^{i1i2} - y_d^{si2} \delta_{si1} - 2 y_d^{i1s} \delta_{si2}) \mathrm{LF}_{3,2,-1}[m_{\tilde e}^{r}, m_{\tilde l}^{p}] +$$
$$\frac{1}{18} \tilde\mu g_1^2 y_e^{pr} \overline{a_e}^{pr} (3 y_d^{i1i2} - y_d^{si2} \delta_{si1} - 2 y_d^{i1s} \delta_{si2}) \mathrm{LF}_{3,1,0}[m_{\tilde l}^{p}, m_{\tilde e}^{r}] +$$
$$\frac{1}{18} \tilde\mu g_1^2 y_e^{pr} \overline{a_e}^{pr} (3 y_d^{i1i2} - y_d^{si2} \delta_{si1} - 2 y_d^{i1s} \delta_{si2}) \mathrm{LF}_{3,2,-1}[m_{\tilde l}^{p}, m_{\tilde e}^{r}] + \frac{1}{8} \tilde\mu y_e^{pr} \overline{a_e}^{pr}$$
$$(-g_2^2 (7 y_d^{i1i2} + y_d^{si2} \delta_{si1}) + g_1^2 (-3 y_d^{i1i2} + y_d^{si2} \delta_{si1} + 2 y_d^{i1s} \delta_{si2})) \mathrm{LF}_{4,1,-1}[m_{\tilde l}^{p}, m_{\tilde e}^{r}] +$$
$$\frac{1}{18} \tilde\mu y_e^{pr} \overline{a_e}^{pr} (3 g_2^2 (6 y_d^{i1i2} + y_d^{si2} \delta_{si1}) + g_1^2 (3 y_d^{i1i2} - y_d^{si2} \delta_{si1} - 2 y_d^{i1s} \delta_{si2}))$$
$$\mathrm{LF}_{5,1,-2}[m_{\tilde l}^{p}, m_{\tilde e}^{r}] + \frac{1}{18} \tilde\mu g_1^2 y_d^{pr} \overline{a_d}^{pr} (3 y_d^{i1i2} - y_d^{si2} \delta_{si1} - 2 y_d^{i1s} \delta_{si2}) \mathrm{LF}_{3,1,0}[m_{\tilde q}^{p}, m_{\tilde d}^{r}] +$$
$$\frac{1}{18} \tilde\mu g_1^2 y_d^{pr} \overline{a_d}^{pr} (3 y_d^{i1i2} - y_d^{si2} \delta_{si1} - 2 y_d^{i1s} \delta_{si2}) \mathrm{LF}_{3,2,-1}[m_{\tilde q}^{p}, m_{\tilde d}^{r}] +$$
$$\frac{1}{24} \tilde\mu y_d^{pr} \overline{a_d}^{pr} (-9 g_2^2 (7 y_d^{i1i2} + y_d^{si2} \delta_{si1}) + 5 g_1^2 (-3 y_d^{i1i2} + y_d^{si2} \delta_{si1} + 2 y_d^{i1s} \delta_{si2}))$$
$$\mathrm{LF}_{4,1,-1}[m_{\tilde q}^{p}, m_{\tilde d}^{r}] + \frac{1}{6} \tilde\mu y_d^{pr} \overline{a_d}^{pr}$$
$$(3 g_2^2 (6 y_d^{i1i2} + y_d^{si2} \delta_{si1}) + g_1^2 (3 y_d^{i1i2} - y_d^{si2} \delta_{si1} - 2 y_d^{i1s} \delta_{si2})) \mathrm{LF}_{5,1,-2}[m_{\tilde q}^{p}, m_{\tilde d}^{r}] +$$
$$\frac{1}{36} \tilde\mu y_u^{pr} \overline{a_u}^{pr} ((g_1^2 + 9 g_2^2) (3 y_d^{i1i2} - y_d^{si2} \delta_{si1}) - 2 g_1^2 y_d^{i1s} \delta_{si2}) \mathrm{LF}_{2,2,0}[m_{\tilde q}^{p}, m_{\tilde u}^{r}] +$$
$$\frac{3}{4} \tilde\mu g_2^2 y_d^{i1i2} y_u^{pr} \overline{a_u}^{pr} \mathrm{LF}_{3,1,0}[m_{\tilde q}^{p}, m_{\tilde u}^{r}] +$$
$$\frac{1}{72} \tilde\mu y_u^{pr} \overline{a_u}^{pr} (-(g_1^2 + 9 g_2^2) (3 y_d^{i1i2} - y_d^{si2} \delta_{si1}) + 2 g_1^2 y_d^{i1s} \delta_{si2}) \mathrm{LF}_{3,2,-1}[m_{\tilde q}^{p}, m_{\tilde u}^{r}] -$$
$$\frac{3}{4} \tilde\mu g_2^2 y_d^{i1i2} y_u^{pr} \overline{a_u}^{pr} \mathrm{LF}_{4,1,-1}[m_{\tilde q}^{p}, m_{\tilde u}^{r}] + \frac{1}{36} \tilde\mu y_u^{pr} \overline{a_u}^{pr}$$
$$(9 g_2^2 (6 y_d^{i1i2} + y_d^{si2} \delta_{si1}) + g_1^2 (-3 y_d^{i1i2} + y_d^{si2} \delta_{si1} + 2 y_d^{i1s} \delta_{si2})) \mathrm{LF}_{3,1,0}[m_{\tilde u}^{r}, m_{\tilde q}^{p}] +$$
$$\frac{1}{36} \tilde\mu y_u^{pr} \overline{a_u}^{pr} (-(g_1^2 + 9 g_2^2) (3 y_d^{i1i2} - y_d^{si2} \delta_{si1}) + 2 g_1^2 y_d^{i1s} \delta_{si2}) \mathrm{LF}_{3,2,-1}[m_{\tilde u}^{r}, m_{\tilde q}^{p}] +$$
$$\frac{1}{12} \tilde\mu y_u^{pr} \overline{a_u}^{pr} (-9 g_2^2 (6 y_d^{i1i2} + y_d^{si2} \delta_{si1}) + g_1^2 (-3 y_d^{i1i2} + y_d^{si2} \delta_{si1} + 2 y_d^{i1s} \delta_{si2}))$$
$$\mathrm{LF}_{4,1,-1}[m_{\tilde u}^{r}, m_{\tilde q}^{p}] + \frac{1}{6} \tilde\mu y_u^{pr} \overline{a_u}^{pr}$$
$$(3 g_2^2 (6 y_d^{i1i2} + y_d^{si2} \delta_{si1}) + g_1^2 (3 y_d^{i1i2} - y_d^{si2} \delta_{si1} - 2 y_d^{i1s} \delta_{si2})) \mathrm{LF}_{5,1,-2}[m_{\tilde u}^{r}, m_{\tilde q}^{p}] +$$
$$\frac{1}{2} m_1 \tilde\mu g_1^2 g_2^2 y_d^{i1i2} \mathrm{LF}_{3,1,0}[\tilde\mu, m_1] + \frac{7}{8} m_1 \tilde\mu g_1^4 y_d^{i1i2} \mathrm{LF}_{3,2,-1}[\tilde\mu, m_1] +$$
$$\frac{1}{24} m_1 \tilde\mu g_1^2 (-3 g_2^2 (12 y_d^{i1i2} + y_d^{pi2} \delta_{pi1}) + g_1^2 (-3 y_d^{i1i2} + y_d^{pi2} \delta_{pi1} + 2 y_d^{i1p} \delta_{pi2}))$$
$$\mathrm{LF}_{4,1,-1}[\tilde\mu, m_1] - \frac{1}{2} m_1 \tilde\mu g_1^4 y_d^{i1i2} \mathrm{LF}_{4,2,-2}[\tilde\mu, m_1] +$$
$$\frac{1}{18} m_1 \tilde\mu g_1^2 (3 g_2^2 (6 y_d^{i1i2} + y_d^{pi2} \delta_{pi1}) + g_1^2 (3 y_d^{i1i2} - y_d^{pi2} \delta_{pi1} - 2 y_d^{i1p} \delta_{pi2}))$$
$$\mathrm{LF}_{5,1,-2}[\tilde\mu, m_1] + \frac{1}{6} m_2 \tilde\mu g_2^4 (9 y_d^{i1i2} + 2 y_d^{pi2} \delta_{pi1}) \mathrm{LF}_{3,1,0}[\tilde\mu, m_2] +$$
$$\frac{1}{24} m_2 \tilde\mu g_2^4 (57 y_d^{i1i2} + 8 y_d^{pi2} \delta_{pi1}) \mathrm{LF}_{3,2,-1}[\tilde\mu, m_2] +$$
$$\frac{1}{8} m_2 \tilde\mu g_2^2 (-g_2^2 (36 y_d^{i1i2} + 7 y_d^{pi2} \delta_{pi1}) + g_1^2 (-3 y_d^{i1i2} + y_d^{pi2} \delta_{pi1} + 2 y_d^{i1p} \delta_{pi2}))$$
$$\mathrm{LF}_{4,1,-1}[\tilde\mu, m_2] - 2 m_2 \tilde\mu g_2^4 y_d^{i1i2} \mathrm{LF}_{4,2,-2}[\tilde\mu, m_2] +$$
$$\frac{1}{6} m_2 \tilde\mu g_2^2 (3 g_2^2 (6 y_d^{i1i2} + y_d^{pi2} \delta_{pi1}) + g_1^2 (3 y_d^{i1i2} - y_d^{pi2} \delta_{pi1} - 2 y_d^{i1p} \delta_{pi2}))$$
$$\mathrm{LF}_{5,1,-2}[\tilde\mu, m_2] + \frac{1}{36} m_1 \tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{2,1,1,0}[m_1, m_{\tilde d}^{i2}, m_{\tilde q}^{p}] -$$
$$\frac{1}{72} m_1 \tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde d}^{i2}, m_{\tilde q}^{p}] -$$
$$\frac{1}{36} m_1 \tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{3,1,1,-1}[m_1, m_{\tilde d}^{i2}, m_{\tilde q}^{p}] +$$
$$\frac{1}{72} m_1 \tilde\mu g_1^2 g_2^2 y_d^{i1i2} \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde d}^{i2}, m_{\tilde q}^{i1}] + \frac{2}{9} m_1 \tilde\mu g_1^4 y_d^{i1i2} \mathrm{LF}_{2,1,1,0}[m_1, m_{\tilde d}^{i2}, \tilde\mu] -$$
$$\frac{1}{9} m_1 \tilde\mu g_1^4 y_d^{i1i2} \mathrm{LF}_{3,1,1,-1}[m_1, m_{\tilde d}^{i2}, \tilde\mu] + \frac{1}{72} m_1 \tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde q}^{p}, m_{\tilde d}^{i2}] +$$
$$\frac{1}{12} m_1 \tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{2,1,1,0}[m_1, m_{\tilde q}^{p}, \tilde\mu] -$$
$$\frac{1}{12} m_1 \tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{3,1,1,-1}[m_1, m_{\tilde q}^{p}, \tilde\mu] +$$
$$\frac{1}{72} m_1 \tilde\mu g_1^2 g_2^2 y_d^{i1i2} \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde q}^{i1}, m_{\tilde d}^{i2}] - \frac{1}{18} m_1 \tilde\mu g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r}$$
$$\mathrm{LF}_{2,1,1,0}[m_1, m_{\tilde q}^{i1}, m_{\tilde u}^{r}] + \frac{1}{36} m_1 \tilde\mu g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde q}^{i1}, m_{\tilde u}^{r}] +$$
$$\frac{1}{18} m_1 \tilde\mu g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{3,1,1,-1}[m_1, m_{\tilde q}^{i1}, m_{\tilde u}^{r}] + \frac{1}{18} m_1 \tilde\mu g_1^4 y_d^{i1i2} \mathrm{LF}_{2,1,1,0}[m_1, m_{\tilde q}^{i1}, \tilde\mu] -$$
$$\frac{1}{36} m_1 \tilde\mu g_1^4 y_d^{i1i2} \mathrm{LF}_{3,1,1,-1}[m_1, m_{\tilde q}^{i1}, \tilde\mu] - \frac{1}{36} m_1 \tilde\mu g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde u}^{r}, m_{\tilde q}^{i1}] +$$
$$\frac{1}{3} m_1 \tilde\mu g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{2,1,1,0}[m_1, m_{\tilde u}^{r}, \tilde\mu] -$$
$$\frac{1}{3} m_1 \tilde\mu g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{3,1,1,-1}[m_1, m_{\tilde u}^{r}, \tilde\mu] -$$
$$\frac{1}{36} m_1 \tilde\mu g_1^2 (3 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} + y_d^{i1i2} (14 g_1^2 + 3 g_2^2)) \mathrm{LF}_{2,2,1,-1}[m_1, \tilde\mu, m_{\tilde d}^{i2}] -$$
$$\frac{1}{6} m_1 \tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{2,2,1,-1}[m_1, \tilde\mu, m_{\tilde q}^{p}] -$$
$$\frac{1}{72} m_1 \tilde\mu g_1^2 (y_d^{i1i2} (13 g_1^2 + 3 g_2^2) - 3 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r}) \mathrm{LF}_{2,2,1,-1}[m_1, \tilde\mu, m_{\tilde q}^{i1}] -$$
$$\frac{7}{24} m_1 \tilde\mu g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{2,2,1,-1}[m_1, \tilde\mu, m_{\tilde u}^{r}] +$$
$$\frac{3}{4} m_2 \tilde\mu g_2^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{2,1,1,0}[m_2, m_{\tilde q}^{p}, \tilde\mu] -$$
$$\frac{3}{4} m_2 \tilde\mu g_2^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{3,1,1,-1}[m_2, m_{\tilde q}^{p}, \tilde\mu] + \frac{3}{2} m_2 \tilde\mu g_2^4 y_d^{i1i2} \mathrm{LF}_{2,1,1,0}[m_2, m_{\tilde q}^{i1}, \tilde\mu] -$$
$$\frac{3}{4} m_2 \tilde\mu g_2^4 y_d^{i1i2} \mathrm{LF}_{3,1,1,-1}[m_2, m_{\tilde q}^{i1}, \tilde\mu] + \frac{3}{2} \tilde\mu g_1^2 g_2^2 y_d^{i1i2} (m_1 + m_2) \mathrm{LF}_{2,2,1,-1}[m_2, \tilde\mu, m_1] -$$
$$\tilde\mu g_1^2 g_2^2 y_d^{i1i2} (m_1 + m_2) \mathrm{LF}_{3,2,1,-2}[m_2, \tilde\mu, m_1] -$$
$$\frac{1}{4} m_2 \tilde\mu g_2^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{2,2,1,-1}[m_2, \tilde\mu, m_{\tilde d}^{r}] +$$
$$\frac{1}{4} m_2 \tilde\mu g_2^2 y_d^{pi2} (-3 \overline{y_d}^{pr} y_d^{i1r} + g_2^2 \delta_{pi1}) \mathrm{LF}_{2,2,1,-1}[m_2, \tilde\mu, m_{\tilde q}^{p}] -$$
$$\frac{1}{8} m_2 \tilde\mu g_2^2 (20 g_2^2 y_d^{i1i2} + 3 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r}) \mathrm{LF}_{2,2,1,-1}[m_2, \tilde\mu, m_{\tilde q}^{i1}] -$$
$$\frac{1}{8} m_2 \tilde\mu g_2^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{2,2,1,-1}[m_2, \tilde\mu, m_{\tilde u}^{r}] -$$
$$\frac{2}{3} m_3 \tilde\mu g_3^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{2,1,1,0}[m_3, m_{\tilde d}^{i2}, m_{\tilde q}^{p}] +$$
$$\frac{1}{3} m_3 \tilde\mu g_3^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{2,2,1,-1}[m_3, m_{\tilde d}^{i2}, m_{\tilde q}^{p}] +$$
$$\frac{2}{3} m_3 \tilde\mu g_3^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{3,1,1,-1}[m_3, m_{\tilde d}^{i2}, m_{\tilde q}^{p}] -$$
$$\frac{1}{3} m_3 \tilde\mu g_2^2 g_3^2 y_d^{i1i2} \mathrm{LF}_{2,2,1,-1}[m_3, m_{\tilde d}^{i2}, m_{\tilde q}^{i1}] -$$
$$\frac{1}{3} m_3 \tilde\mu g_3^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{2,2,1,-1}[m_3, m_{\tilde q}^{p}, m_{\tilde d}^{i2}] -$$
$$\frac{1}{3} m_3 \tilde\mu g_2^2 g_3^2 y_d^{i1i2} \mathrm{LF}_{2,2,1,-1}[m_3, m_{\tilde q}^{i1}, m_{\tilde d}^{i2}] - \frac{2}{3} m_3 \tilde\mu g_3^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r}$$
$$\mathrm{LF}_{2,1,1,0}[m_3, m_{\tilde q}^{i1}, m_{\tilde u}^{r}] + \frac{1}{3} m_3 \tilde\mu g_3^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{2,2,1,-1}[m_3, m_{\tilde q}^{i1}, m_{\tilde u}^{r}] +$$
$$\frac{2}{3} m_3 \tilde\mu g_3^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{3,1,1,-1}[m_3, m_{\tilde q}^{i1}, m_{\tilde u}^{r}] - \frac{1}{3} m_3 \tilde\mu g_3^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r}$$
$$\mathrm{LF}_{2,2,1,-1}[m_3, m_{\tilde u}^{r}, m_{\tilde q}^{i1}] + \frac{1}{8} \tilde\mu y_d^{si2} y_d^{i1r} \overline{y_u}^{st} y_u^{pt} \overline{a_d}^{pr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde d}^{r}, \tilde\mu, m_{\tilde q}^{p}] -$$
$$\frac{1}{36} m_1 \tilde\mu g_1^2 g_2^2 y_d^{i1i2} \mathrm{LF}_{2,1,1,0}[m_{\tilde d}^{i2}, m_1, m_{\tilde q}^{i1}] + \frac{1}{36} m_1 \tilde\mu g_1^2 g_2^2 y_d^{i1i2} \mathrm{LF}_{3,1,1,-1}[m_{\tilde d}^{i2}, m_1, m_{\tilde q}^{i1}] +$$
$$\frac{2}{3} m_3 \tilde\mu g_2^2 g_3^2 y_d^{i1i2} \mathrm{LF}_{2,1,1,0}[m_{\tilde d}^{i2}, m_3, m_{\tilde q}^{i1}] - \frac{2}{3} m_3 \tilde\mu g_2^2 g_3^2 y_d^{i1i2} \mathrm{LF}_{3,1,1,-1}[m_{\tilde d}^{i2}, m_3, m_{\tilde q}^{i1}] +$$
$$\tilde\mu y_d^{i1i2} \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde l}^{p}, m_{\tilde e}^{r}, m_{\tilde l}^{s}] -$$
$$\tilde\mu y_d^{i1i2} \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde l}^{p}, m_{\tilde e}^{r}, m_{\tilde l}^{s}] -$$
$$\frac{1}{4} \tilde\mu y_d^{i1i2} \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde l}^{p}, m_{\tilde l}^{s}, m_{\tilde e}^{r}] +$$
$$3 \tilde\mu \overline{y_d}^{st} y_d^{pt} y_d^{sr} y_d^{i1i2} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{p}, m_{\tilde d}^{r}, m_{\tilde q}^{s}] -$$
$$3 \tilde\mu \overline{y_d}^{st} y_d^{pt} y_d^{sr} y_d^{i1i2} \overline{a_d}^{pr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde d}^{r}, m_{\tilde q}^{s}] +$$
$$3 \tilde\mu y_d^{sr} y_d^{i1i2} \overline{y_u}^{st} y_u^{pt} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{p}, m_{\tilde d}^{r}, m_{\tilde u}^{t}] -$$
$$3 \tilde\mu y_d^{sr} y_d^{i1i2} \overline{y_u}^{st} y_u^{pt} \overline{a_d}^{pr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde d}^{r}, m_{\tilde u}^{t}] -$$
$$\frac{3}{4} \tilde\mu \overline{y_d}^{st} y_d^{pt} y_d^{sr} y_d^{i1i2} \overline{a_d}^{pr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde d}^{r}] -$$
$$\frac{3}{4} \tilde\mu \overline{y_d}^{st} y_d^{pt} y_d^{i1i2} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{r}] +$$
$$\frac{3}{4} \tilde\mu \overline{y_d}^{st} y_d^{pt} y_d^{i1i2} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{r}] +$$
$$\frac{9}{8} \tilde\mu \overline{y_d}^{st} y_d^{pt} y_d^{i1i2} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^{p}, m_{\tilde u}^{r}, m_{\tilde q}^{s}] +$$
$$\frac{3}{4} \tilde\mu g_2^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{p}, m_{\tilde u}^{r}, \tilde\mu] - \frac{1}{4} \tilde\mu g_2^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde u}^{r}, \tilde\mu] -$$
$$3 \tilde\mu y_d^{sr} y_d^{i1i2} \overline{y_u}^{st} y_u^{pt} \overline{a_d}^{pr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^{p}, m_{\tilde u}^{t}, m_{\tilde d}^{r}] -$$
$$\frac{1}{8} \tilde\mu y_d^{si2} y_d^{i1r} \overline{y_u}^{st} y_u^{pt} \overline{a_d}^{pr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^{p}, \tilde\mu, m_{\tilde d}^{r}] +$$
$$\frac{1}{8} \tilde\mu y_d^{pi2} \overline{a_u}^{pr} (\overline{y_d}^{st} y_d^{i1t} y_u^{sr} - g_2^2 y_u^{i1r}) \mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^{p}, \tilde\mu, m_{\tilde u}^{r}] -$$
$$\frac{3}{4} \tilde\mu \overline{y_d}^{st} y_d^{pt} y_d^{i1i2} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{s}, m_{\tilde q}^{p}, m_{\tilde u}^{r}] +$$
$$\frac{3}{4} \tilde\mu \overline{y_d}^{st} y_d^{pt} y_d^{i1i2} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{s}, m_{\tilde q}^{p}, m_{\tilde u}^{r}] -$$
$$\frac{3}{8} \tilde\mu \overline{y_d}^{st} y_d^{pt} y_d^{i1i2} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^{s}, m_{\tilde u}^{r}, m_{\tilde q}^{p}] +$$
$$\frac{1}{36} m_1 \tilde\mu g_1^2 g_2^2 y_d^{i1i2} \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{i1}, m_1, m_{\tilde d}^{i2}] + \frac{1}{36} m_1 \tilde\mu g_1^2 g_2^2 y_d^{i1i2} \mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{i1}, m_1, m_{\tilde d}^{i2}] +$$
$$\frac{1}{12} m_1 \tilde\mu g_1^2 g_2^2 y_d^{i1i2} \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{i1}, m_1, \tilde\mu] - \frac{1}{4} m_2 \tilde\mu g_2^4 y_d^{i1i2} \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{i1}, m_2, \tilde\mu] -$$
$$\frac{2}{3} m_3 \tilde\mu g_2^2 g_3^2 y_d^{i1i2} \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{i1}, m_3, m_{\tilde d}^{i2}] - \frac{2}{3} m_3 \tilde\mu g_2^2 g_3^2 y_d^{i1i2} \mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{i1}, m_3, m_{\tilde d}^{i2}] -$$
$$\frac{3}{2} \tilde\mu \overline{y_d}^{st} y_d^{pt} y_d^{i1i2} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde u}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}] +$$
$$\frac{3}{2} \tilde\mu \overline{y_d}^{st} y_d^{pt} y_d^{i1i2} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde u}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}] +$$
$$\frac{1}{4} \tilde\mu g_2^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde u}^{r}, m_{\tilde q}^{p}, \tilde\mu] - \frac{1}{4} \tilde\mu g_2^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde u}^{r}, m_{\tilde q}^{p}, \tilde\mu] -$$
$$\frac{1}{8} \tilde\mu y_d^{pi2} \overline{a_u}^{pr} (\overline{y_d}^{st} y_d^{i1t} y_u^{sr} + g_2^2 y_u^{i1r}) \mathrm{LF}_{2,2,1,-1}[m_{\tilde u}^{r}, \tilde\mu, m_{\tilde q}^{p}] -$$
$$\frac{1}{4} \tilde\mu g_1^2 g_2^2 y_d^{i1i2} (m_1 - 5 m_2) \mathrm{LF}_{3,1,1,-1}[\tilde\mu, m_1, m_2] -$$
$$\frac{3}{2} m_2 \tilde\mu g_1^2 g_2^2 y_d^{i1i2} \mathrm{LF}_{4,1,1,-2}[\tilde\mu, m_1, m_2] + \frac{1}{6} m_1 \tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{2,1,1,0}[\tilde\mu, m_1, m_{\tilde d}^{i2}] -$$
$$\frac{1}{6} m_1 \tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{3,1,1,-1}[\tilde\mu, m_1, m_{\tilde d}^{i2}] +$$
$$\frac{1}{2} m_1 \tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{2,1,1,0}[\tilde\mu, m_1, m_{\tilde q}^{p}] -$$
$$\frac{1}{4} m_1 \tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{3,1,1,-1}[\tilde\mu, m_1, m_{\tilde q}^{p}] - \frac{1}{12} m_1 \tilde\mu g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r}$$
$$\mathrm{LF}_{2,1,1,0}[\tilde\mu, m_1, m_{\tilde q}^{i1}] + \frac{1}{12} m_1 \tilde\mu g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{3,1,1,-1}[\tilde\mu, m_1, m_{\tilde q}^{i1}] +$$
$$\frac{1}{2} m_1 \tilde\mu g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{2,1,1,0}[\tilde\mu, m_1, m_{\tilde u}^{r}] - \frac{1}{4} m_1 \tilde\mu g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r}$$
$$\mathrm{LF}_{3,1,1,-1}[\tilde\mu, m_1, m_{\tilde u}^{r}] - \frac{1}{2} \tilde\mu g_1^2 g_2^2 y_d^{i1i2} (m_1 + 3 m_2) \mathrm{LF}_{3,2,1,-2}[\tilde\mu, m_2, m_1] +$$
$$m_2 \tilde\mu g_2^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{2,1,1,0}[\tilde\mu, m_2, m_{\tilde d}^{r}] - \frac{1}{2} m_2 \tilde\mu g_2^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{3,1,1,-1}[\tilde\mu, m_2, m_{\tilde d}^{r}] +$$
$$\frac{3}{2} m_2 \tilde\mu g_2^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{2,1,1,0}[\tilde\mu, m_2, m_{\tilde q}^{p}] - \frac{3}{4} m_2 \tilde\mu g_2^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r}$$
$$\mathrm{LF}_{3,1,1,-1}[\tilde\mu, m_2, m_{\tilde q}^{p}] + \frac{3}{4} m_2 \tilde\mu g_2^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{2,1,1,0}[\tilde\mu, m_2, m_{\tilde q}^{i1}] -$$
$$\frac{3}{4} m_2 \tilde\mu g_2^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{3,1,1,-1}[\tilde\mu, m_2, m_{\tilde q}^{i1}] + \frac{1}{2} m_2 \tilde\mu g_2^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r}$$
$$\mathrm{LF}_{2,1,1,0}[\tilde\mu, m_2, m_{\tilde u}^{r}] - \frac{1}{4} m_2 \tilde\mu g_2^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{3,1,1,-1}[\tilde\mu, m_2, m_{\tilde u}^{r}] -$$
$$\frac{1}{4} \tilde\mu y_d^{si2} y_d^{i1r} \overline{y_u}^{st} y_u^{pt} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,0}[\tilde\mu, m_{\tilde d}^{r}, m_{\tilde q}^{p}] + \frac{1}{4} \tilde\mu y_d^{si2} y_d^{i1r} \overline{y_u}^{st} y_u^{pt} \overline{a_d}^{pr}$$
$$\mathrm{LF}_{3,1,1,-1}[\tilde\mu, m_{\tilde d}^{r}, m_{\tilde q}^{p}] - \frac{1}{4} \tilde\mu \overline{y_d}^{st} y_d^{pi2} y_d^{i1t} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,0}[\tilde\mu, m_{\tilde q}^{p}, m_{\tilde u}^{r}] +$$
$$\frac{1}{4} \tilde\mu \overline{y_d}^{st} y_d^{pi2} y_d^{i1t} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{3,1,1,-1}[\tilde\mu, m_{\tilde q}^{p}, m_{\tilde u}^{r}] - \frac{1}{6} \tilde\mu g_1^2 g_2^2 y_d^{i1i2} (m_1 + m_2)$$
$$\mathrm{LF}_{1,1,1,1,0}[m_1, m_2, m_{\tilde q}^{i1}, \tilde\mu] + \frac{1}{12} m_2 \tilde\mu g_1^2 g_2^2 y_d^{i1i2} \mathrm{LF}_{2,1,1,1,-1}[m_1, m_2, m_{\tilde q}^{i1}, \tilde\mu] -$$
$$\frac{1}{4} \tilde\mu g_1^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde d}^{r}, m_{\tilde q}^{p}, \tilde\mu] + \frac{1}{8} \tilde\mu g_1^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr}$$
$$\mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde d}^{r}, m_{\tilde q}^{p}, \tilde\mu] - \frac{1}{9} \tilde\mu g_1^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde d}^{i2}, m_{\tilde d}^{r}, m_{\tilde q}^{p}] +$$
$$\frac{1}{18} \tilde\mu g_1^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde d}^{i2}, m_{\tilde d}^{r}, m_{\tilde q}^{p}] + \frac{1}{4} m_1 \tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r}$$
$$\mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde d}^{i2}, m_{\tilde q}^{p}, \tilde\mu] - \frac{1}{24} m_1 \tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde d}^{i2}, m_{\tilde q}^{p}, \tilde\mu] -$$
$$\frac{1}{9} m_1 \tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde d}^{i2}, m_{\tilde q}^{i1}, m_{\tilde q}^{p}] -$$
$$\frac{1}{9} m_1 \tilde\mu g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde d}^{i2}, m_{\tilde q}^{i1}, m_{\tilde u}^{r}] +$$
$$\frac{1}{9} \tilde\mu g_1^4 m_1^2 a_d^{i1i2} \mathrm{LF}_{2,1,1,1,0}[m_1, m_{\tilde d}^{i2}, m_{\tilde q}^{i1}, \tilde\mu] +$$
$$\frac{1}{3} m_1 \tilde\mu g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde d}^{i2}, m_{\tilde u}^{r}, \tilde\mu] +$$
$$\frac{1}{8} \tilde\mu g_1^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde q}^{p}, m_{\tilde u}^{r}, \tilde\mu] -$$
$$\frac{1}{16} \tilde\mu g_1^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde q}^{p}, m_{\tilde u}^{r}, \tilde\mu] -$$
$$\frac{1}{36} \tilde\mu g_1^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, m_{\tilde u}^{r}] +$$
$$\frac{1}{72} \tilde\mu g_1^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, m_{\tilde u}^{r}] -$$
$$\frac{1}{6} m_1 \tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, \tilde\mu] -$$
$$\frac{1}{8} m_1 \tilde\mu g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde q}^{i1}, m_{\tilde u}^{r}, \tilde\mu] +$$
$$\frac{1}{48} m_1 \tilde\mu g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde q}^{i1}, m_{\tilde u}^{r}, \tilde\mu] +$$
$$\frac{1}{12} m_1 \tilde\mu g_1^2 g_2^2 y_d^{i1i2} \mathrm{LF}_{2,1,1,1,-1}[m_2, m_1, m_{\tilde q}^{i1}, \tilde\mu] -$$
$$\frac{3}{4} \tilde\mu g_2^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde d}^{r}, m_{\tilde q}^{p}, \tilde\mu] +$$
$$\frac{3}{8} \tilde\mu g_2^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_2, m_{\tilde d}^{r}, m_{\tilde q}^{p}, \tilde\mu] -$$
$$\frac{1}{2} \tilde\mu g_2^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde d}^{r}, m_{\tilde q}^{i1}, m_{\tilde q}^{p}] +$$
$$\frac{1}{4} \tilde\mu g_2^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_2, m_{\tilde d}^{r}, m_{\tilde q}^{i1}, m_{\tilde q}^{p}] +$$
$$\frac{3}{4} m_2 \tilde\mu g_2^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde d}^{r}, m_{\tilde q}^{i1}, \tilde\mu] -$$
$$\frac{1}{8} m_2 \tilde\mu g_2^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{2,1,1,1,-1}[m_2, m_{\tilde d}^{r}, m_{\tilde q}^{i1}, \tilde\mu] -$$
$$\frac{3}{8} \tilde\mu g_2^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde q}^{p}, m_{\tilde u}^{r}, \tilde\mu] +$$
$$\frac{3}{16} \tilde\mu g_2^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_2, m_{\tilde q}^{p}, m_{\tilde u}^{r}, \tilde\mu] -$$
$$\frac{1}{4} \tilde\mu g_2^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, m_{\tilde u}^{r}] +$$
$$\frac{1}{8} \tilde\mu g_2^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_2, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, m_{\tilde u}^{r}] +$$
$$\frac{1}{2} m_2 \tilde\mu g_2^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, \tilde\mu] +$$
$$\frac{3}{8} m_2 \tilde\mu g_2^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde q}^{i1}, m_{\tilde u}^{r}, \tilde\mu] -$$
$$\frac{1}{16} m_2 \tilde\mu g_2^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}[m_2, m_{\tilde q}^{i1}, m_{\tilde u}^{r}, \tilde\mu] -$$
$$\frac{4}{3} \tilde\mu g_3^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_3, m_{\tilde d}^{i2}, m_{\tilde d}^{r}, m_{\tilde q}^{p}] +$$
$$\frac{2}{3} \tilde\mu g_3^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_3, m_{\tilde d}^{i2}, m_{\tilde d}^{r}, m_{\tilde q}^{p}] +$$
$$\frac{8}{3} m_3 \tilde\mu g_3^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{1,1,1,1,0}[m_3, m_{\tilde d}^{i2}, m_{\tilde q}^{i1}, m_{\tilde q}^{p}] +$$
$$\frac{8}{3} m_3 \tilde\mu g_3^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{1,1,1,1,0}[m_3, m_{\tilde d}^{i2}, m_{\tilde q}^{i1}, m_{\tilde u}^{r}] -$$
$$\frac{4}{3} \tilde\mu g_3^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_3, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, m_{\tilde u}^{r}] +$$
$$\frac{2}{3} \tilde\mu g_3^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_3, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, m_{\tilde u}^{r}] -$$
$$\frac{1}{18} \tilde\mu g_1^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde d}^{r}, m_1, m_{\tilde d}^{i2}, m_{\tilde q}^{p}] -$$
$$\frac{1}{24} \tilde\mu g_1^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde d}^{r}, m_1, m_{\tilde q}^{p}, \tilde\mu] +$$
$$\frac{1}{8} \tilde\mu g_2^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde d}^{r}, m_2, m_{\tilde q}^{p}, \tilde\mu] +$$
$$\frac{1}{8} m_2 \tilde\mu g_2^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde d}^{r}, m_2, m_{\tilde q}^{i1}, \tilde\mu] -$$
$$\frac{2}{3} \tilde\mu g_3^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde d}^{r}, m_3, m_{\tilde d}^{i2}, m_{\tilde q}^{p}] -$$
$$\frac{1}{2} \tilde\mu \overline{y_d}^{st} y_d^{pt} y_d^{si2} y_d^{i1r} \overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{p}, \tilde\mu] +$$
$$\frac{1}{4} \tilde\mu \overline{y_d}^{st} y_d^{pt} y_d^{si2} y_d^{i1r} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde d}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{p}, \tilde\mu] +$$
$$\frac{3}{4} \tilde\mu y_d^{sr} y_d^{i1i2} \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{pt} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde d}^{r}, m_{\tilde q}^{p}, m_{\tilde d}^{t}, m_{\tilde q}^{s}] -$$
$$\frac{1}{2} \tilde\mu \overline{y_d}^{st} y_d^{pi2} y_d^{sr} y_d^{i1t} \overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, \tilde\mu] -$$
$$\tilde\mu y_d^{pi2} y_d^{sr} \overline{y_u}^{st} y_u^{i1t} \overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde q}^{p}, m_{\tilde u}^{t}, \tilde\mu] -$$
$$\frac{1}{4} \tilde\mu \overline{y_d}^{st} y_d^{pt} y_d^{si2} y_d^{i1r} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde d}^{t}, m_{\tilde d}^{r}, m_{\tilde q}^{p}, \tilde\mu] -$$
$$\frac{3}{4} \tilde\mu y_d^{sr} y_d^{i1i2} \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{pt} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde d}^{t}, m_{\tilde q}^{p}, m_{\tilde d}^{r}, m_{\tilde q}^{s}] +$$
$$\frac{1}{18} \tilde\mu g_1^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde d}^{i2}, m_1, m_{\tilde d}^{r}, m_{\tilde q}^{p}] -$$
$$\frac{1}{24} m_1 \tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde d}^{i2}, m_1, m_{\tilde q}^{p}, \tilde\mu] +$$
$$\frac{2}{3} \tilde\mu g_3^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde d}^{i2}, m_3, m_{\tilde d}^{r}, m_{\tilde q}^{p}] +$$
$$\frac{1}{4} \tilde\mu y_d^{i1i2} y_e^{sr} \overline{a_e}^{pr} \overline{a_e}^{st} a_e^{pt} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde e}^{r}, m_{\tilde l}^{p}, m_{\tilde e}^{t}, m_{\tilde l}^{s}] -$$
$$\frac{1}{4} \tilde\mu y_d^{i1i2} y_e^{sr} \overline{a_e}^{pr} \overline{a_e}^{st} a_e^{pt} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde e}^{t}, m_{\tilde l}^{p}, m_{\tilde e}^{r}, m_{\tilde l}^{s}] +$$
$$\frac{1}{2} \tilde\mu y_d^{i1i2} y_e^{sr} \overline{a_e}^{pr} \overline{a_e}^{st} a_e^{pt} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde l}^{p}, m_{\tilde e}^{r}, m_{\tilde e}^{t}, m_{\tilde l}^{s}] -$$
$$\frac{1}{2} \tilde\mu y_d^{i1i2} y_e^{sr} \overline{a_e}^{pr} \overline{a_e}^{st} a_e^{pt} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde l}^{p}, m_{\tilde e}^{r}, m_{\tilde e}^{t}, m_{\tilde l}^{s}] +$$
$$\frac{1}{24} \tilde\mu g_1^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{p}, m_1, m_{\tilde d}^{r}, \tilde\mu] +$$
$$\frac{1}{24} m_1 \tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{p}, m_1, m_{\tilde d}^{i2}, \tilde\mu] -$$
$$\frac{1}{72} \tilde\mu g_1^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{p}, m_1, m_{\tilde q}^{i1}, m_{\tilde u}^{r}] +$$
$$\frac{1}{48} \tilde\mu g_1^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{p}, m_1, m_{\tilde u}^{r}, \tilde\mu] -$$
$$\frac{1}{4} \tilde\mu g_2^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{p}, m_2, m_{\tilde d}^{r}, m_{\tilde q}^{i1}] -$$
$$\frac{1}{8} \tilde\mu g_2^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{p}, m_2, m_{\tilde d}^{r}, \tilde\mu] -$$
$$\frac{1}{8} \tilde\mu g_2^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{p}, m_2, m_{\tilde q}^{i1}, m_{\tilde u}^{r}] -$$
$$\frac{1}{16} \tilde\mu g_2^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{p}, m_2, m_{\tilde u}^{r}, \tilde\mu] -$$
$$\frac{2}{3} \tilde\mu g_3^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{p}, m_3, m_{\tilde q}^{i1}, m_{\tilde u}^{r}] +$$
$$\frac{3}{2} \tilde\mu y_d^{sr} y_d^{i1i2} \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{pt} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^{p}, m_{\tilde d}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{s}] -$$
$$\frac{3}{2} \tilde\mu y_d^{sr} y_d^{i1i2} \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{pt} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde d}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{s}] +$$
$$3 \tilde\mu y_d^{i1i2} y_u^{pt} \overline{a_d}^{pr} (a_d^{sr} \overline{a_u}^{st} - \tilde\mu^2 y_d^{sr} \overline{y_u}^{st}) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^{p}, m_{\tilde d}^{r}, m_{\tilde q}^{s}, m_{\tilde u}^{t}] +$$
$$3 \tilde\mu y_d^{i1i2} y_u^{pt} \overline{a_d}^{pr} (-a_d^{sr} \overline{a_u}^{st} + \tilde\mu^2 y_d^{sr} \overline{y_u}^{st}) \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde d}^{r}, m_{\tilde q}^{s}, m_{\tilde u}^{t}] +$$
$$\frac{1}{4} \tilde\mu \overline{y_d}^{st} y_d^{pi2} y_d^{sr} y_d^{i1t} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde d}^{r}, m_{\tilde q}^{s}, \tilde\mu] +$$
$$\frac{3}{4} \tilde\mu y_d^{i1i2} y_u^{pt} \overline{a_d}^{pr} (-3 a_d^{sr} \overline{a_u}^{st} + \tilde\mu^2 y_d^{sr} \overline{y_u}^{st}) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde d}^{r}, m_{\tilde u}^{t}] +$$
$$\frac{3}{4} \tilde\mu^3 y_d^{i1i2} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{r}, m_{\tilde u}^{t}] -$$
$$\frac{3}{4} \tilde\mu^3 y_d^{i1i2} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{r}, m_{\tilde u}^{t}] -$$
$$\frac{1}{2} \tilde\mu \overline{y_d}^{st} \overline{a_u}^{pr} (y_d^{pi2} y_d^{i1t} y_u^{sr} + 2 y_d^{pt} y_d^{si2} y_u^{i1r}) \mathrm{LF}_{1,1,1,1,0}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{r}, \tilde\mu] +$$
$$\frac{1}{4} \tilde\mu \overline{y_d}^{st} y_d^{pi2} y_d^{i1t} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{r}, \tilde\mu] -$$
$$\frac{9}{8} \tilde\mu^3 y_d^{i1i2} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde u}^{r}, m_{\tilde q}^{s}, m_{\tilde u}^{t}] -$$
$$\frac{1}{2} \tilde\mu y_d^{si2} \overline{y_u}^{st} y_u^{pt} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_{\tilde q}^{p}, m_{\tilde u}^{r}, m_{\tilde u}^{t}, \tilde\mu] +$$
$$3 \tilde\mu y_d^{i1i2} y_u^{pt} \overline{a_d}^{pr} (-a_d^{sr} \overline{a_u}^{st} + \tilde\mu^2 y_d^{sr} \overline{y_u}^{st}) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde u}^{t}, m_{\tilde d}^{r}, m_{\tilde q}^{s}] -$$
$$\frac{3}{8} \tilde\mu^3 y_d^{i1i2} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde u}^{t}, m_{\tilde q}^{s}, m_{\tilde u}^{r}] -$$
$$\frac{1}{4} \tilde\mu \overline{y_d}^{st} y_d^{pi2} y_d^{sr} y_d^{i1t} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{s}, m_{\tilde d}^{r}, m_{\tilde q}^{p}, \tilde\mu] +$$
$$\frac{3}{4} \tilde\mu^3 y_d^{i1i2} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^{s}, m_{\tilde q}^{p}, m_{\tilde u}^{r}, m_{\tilde u}^{t}] -$$
$$\frac{3}{4} \tilde\mu^3 y_d^{i1i2} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^{s}, m_{\tilde q}^{p}, m_{\tilde u}^{r}, m_{\tilde u}^{t}] -$$
$$\frac{1}{4} \tilde\mu \overline{y_d}^{st} y_d^{pi2} y_d^{i1t} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{s}, m_{\tilde q}^{p}, m_{\tilde u}^{r}, \tilde\mu] +$$
$$\frac{3}{8} \tilde\mu^3 y_d^{i1i2} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{s}, m_{\tilde u}^{r}, m_{\tilde q}^{p}, m_{\tilde u}^{t}] -$$
$$\frac{3}{8} \tilde\mu^3 y_d^{i1i2} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{s}, m_{\tilde u}^{t}, m_{\tilde q}^{p}, m_{\tilde u}^{r}] +$$
$$\frac{1}{72} \tilde\mu g_1^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{i1}, m_1, m_{\tilde q}^{p}, m_{\tilde u}^{r}] +$$
$$\frac{1}{48} m_1 \tilde\mu g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{i1}, m_1, m_{\tilde u}^{r}, \tilde\mu] +$$
$$\frac{1}{4} \tilde\mu g_2^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{i1}, m_2, m_{\tilde d}^{r}, m_{\tilde q}^{p}] -$$
$$\frac{1}{8} m_2 \tilde\mu g_2^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{i1}, m_2, m_{\tilde d}^{r}, \tilde\mu] +$$
$$\frac{1}{8} \tilde\mu g_2^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{i1}, m_2, m_{\tilde q}^{p}, m_{\tilde u}^{r}] -$$
$$\frac{1}{16} m_2 \tilde\mu g_2^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{i1}, m_2, m_{\tilde u}^{r}, \tilde\mu] +$$
$$\frac{2}{3} \tilde\mu g_3^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{i1}, m_3, m_{\tilde q}^{p}, m_{\tilde u}^{r}] -$$
$$\frac{1}{48} \tilde\mu g_1^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde u}^{r}, m_1, m_{\tilde q}^{p}, \tilde\mu] -$$
$$\frac{1}{48} m_1 \tilde\mu g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde u}^{r}, m_1, m_{\tilde q}^{i1}, \tilde\mu] +$$
$$\frac{1}{16} \tilde\mu g_2^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde u}^{r}, m_2, m_{\tilde q}^{p}, \tilde\mu] +$$
$$\frac{1}{16} m_2 \tilde\mu g_2^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde u}^{r}, m_2, m_{\tilde q}^{i1}, \tilde\mu] +$$
$$\frac{3}{2} \tilde\mu^3 y_d^{i1i2} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde u}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{t}] -$$
$$\frac{3}{2} \tilde\mu^3 y_d^{i1i2} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde u}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{t}] +$$
$$\frac{1}{4} \tilde\mu y_d^{si2} \overline{y_u}^{st} y_u^{pt} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde u}^{r}, m_{\tilde q}^{p}, m_{\tilde u}^{t}, \tilde\mu] -$$
$$\frac{3}{4} \tilde\mu^3 y_d^{i1i2} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde u}^{r}, m_{\tilde u}^{t}, m_{\tilde q}^{p}, m_{\tilde q}^{s}] -$$
$$\frac{1}{4} \tilde\mu y_d^{si2} \overline{y_u}^{st} y_u^{pt} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde u}^{t}, m_{\tilde q}^{p}, m_{\tilde u}^{r}, \tilde\mu] -$$
$$\frac{1}{3} m_2 \tilde\mu g_1^2 g_2^2 y_d^{i1i2} \mathrm{LF}_{2,1,1,1,-1}[\tilde\mu, m_1, m_2, m_{\tilde d}^{i2}] -$$
$$\frac{1}{24} \tilde\mu g_1^2 g_2^2 y_d^{i1i2} (11 m_1 + 3 m_2) \mathrm{LF}_{2,1,1,1,-1}[\tilde\mu, m_1, m_2, m_{\tilde q}^{i1}] +$$
$$\frac{13}{24} \tilde\mu g_1^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[\tilde\mu, m_1, m_{\tilde d}^{r}, m_{\tilde q}^{p}] -$$
$$\frac{1}{8} m_1 \tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{2,1,1,1,-1}[\tilde\mu, m_1, m_{\tilde d}^{i2}, m_{\tilde q}^{p}] -$$
$$\frac{25}{48} \tilde\mu g_1^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[\tilde\mu, m_1, m_{\tilde q}^{p}, m_{\tilde u}^{r}] -$$
$$\frac{1}{2} m_1 g_1^2 \tilde\mu^3 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{2,1,1,1,0}[\tilde\mu, m_1, m_{\tilde q}^{p}, m_{\tilde u}^{r}] +$$
$$\frac{1}{16} m_1 \tilde\mu g_1^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}[\tilde\mu, m_1, m_{\tilde q}^{i1}, m_{\tilde u}^{r}] +$$
$$\frac{5}{8} \tilde\mu g_2^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[\tilde\mu, m_2, m_{\tilde d}^{r}, m_{\tilde q}^{p}] -$$
$$\frac{3}{8} m_2 \tilde\mu g_2^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} \mathrm{LF}_{2,1,1,1,-1}[\tilde\mu, m_2, m_{\tilde d}^{r}, m_{\tilde q}^{i1}] +$$
$$\frac{9}{16} \tilde\mu g_2^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[\tilde\mu, m_2, m_{\tilde q}^{p}, m_{\tilde u}^{r}] -$$
$$\frac{1}{2} m_2 g_2^2 \tilde\mu^3 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{2,1,1,1,0}[\tilde\mu, m_2, m_{\tilde q}^{p}, m_{\tilde u}^{r}] -$$
$$\frac{3}{16} m_2 \tilde\mu g_2^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}[\tilde\mu, m_2, m_{\tilde q}^{i1}, m_{\tilde u}^{r}] +$$
$$\frac{1}{4} \tilde\mu \overline{y_d}^{st} y_d^{pt} y_d^{si2} y_d^{i1r} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[\tilde\mu, m_{\tilde d}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{p}] +$$
$$\frac{1}{4} \tilde\mu \overline{y_d}^{st} y_d^{pi2} y_d^{sr} y_d^{i1t} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[\tilde\mu, m_{\tilde d}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}] +$$
$$\frac{1}{4} \tilde\mu \overline{y_d}^{st} y_d^{pi2} y_d^{i1t} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[\tilde\mu, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{r}] +$$
$$\frac{1}{4} \tilde\mu y_d^{si2} \overline{y_u}^{st} y_u^{pt} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[\tilde\mu, m_{\tilde q}^{p}, m_{\tilde u}^{r}, m_{\tilde u}^{t}] -$$
$$\frac{1}{3} \tilde\mu g_1^2 g_2^2 y_d^{i1i2} (m_1 + m_2) \mathrm{LF}_{1,1,1,1,1,-1}[m_1, m_2, m_{\tilde d}^{i2}, m_{\tilde q}^{i1}, \tilde\mu] -$$
$$\frac{1}{3} m_1 m_2 \tilde\mu g_1^2 g_2^2 a_d^{i1i2} \mathrm{LF}_{1,1,1,1,1,0}[m_1, m_2, m_{\tilde d}^{i2}, m_{\tilde q}^{i1}, \tilde\mu] -$$
$$\frac{1}{6} m_1 \tilde\mu g_1^2 y_d^{pi2} \overline{a_d}^{pr} a_d^{i1r} \mathrm{LF}_{1,1,1,1,1,0}[m_1, m_{\tilde d}^{r}, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, \tilde\mu] +$$
$$\frac{1}{3} \tilde\mu g_1^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,1,-1}[m_1, m_{\tilde d}^{i2}, m_{\tilde d}^{r}, m_{\tilde q}^{p}, \tilde\mu] -$$
$$\frac{1}{9} m_1 \tilde\mu g_1^2 y_d^{pi2} \overline{a_d}^{pr} a_d^{i1r} \mathrm{LF}_{1,1,1,1,1,0}[m_1, m_{\tilde d}^{i2}, m_{\tilde d}^{r}, m_{\tilde q}^{i1}, m_{\tilde q}^{p}] +$$
$$\frac{1}{3} m_1 \tilde\mu g_1^2 y_u^{i1r} (a_d^{pi2} \overline{a_u}^{pr} - \tilde\mu^2 y_d^{pi2} \overline{y_u}^{pr}) \mathrm{LF}_{1,1,1,1,1,0}[m_1, m_{\tilde d}^{i2}, m_{\tilde q}^{p}, m_{\tilde u}^{r}, \tilde\mu] +$$
$$\frac{1}{9} m_1 \tilde\mu g_1^2 y_u^{i1r} (-a_d^{pi2} \overline{a_u}^{pr} + \tilde\mu^2 y_d^{pi2} \overline{y_u}^{pr}) \mathrm{LF}_{1,1,1,1,1,0}[m_1, m_{\tilde d}^{i2}, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, m_{\tilde u}^{r}] +$$
$$\frac{1}{6} \tilde\mu g_1^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,1,-1}[m_1, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, m_{\tilde u}^{r}, \tilde\mu] +$$
$$\frac{1}{6} m_1 g_1^2 \tilde\mu^3 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{1,1,1,1,1,0}[m_1, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, m_{\tilde u}^{r}, \tilde\mu] +$$
$$\tilde\mu g_2^2 y_d^{pi2} y_d^{i1r} \overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,1,-1}[m_2, m_{\tilde d}^{r}, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, \tilde\mu] +$$
$$\frac{1}{2} m_2 \tilde\mu g_2^2 y_d^{pi2} \overline{a_d}^{pr} a_d^{i1r} \mathrm{LF}_{1,1,1,1,1,0}[m_2, m_{\tilde d}^{r}, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, \tilde\mu] +$$
$$\frac{1}{2} \tilde\mu g_2^2 y_d^{pi2} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,1,-1}[m_2, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, m_{\tilde u}^{r}, \tilde\mu] -$$
$$\frac{1}{2} m_2 g_2^2 \tilde\mu^3 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \mathrm{LF}_{1,1,1,1,1,0}[m_2, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, m_{\tilde u}^{r}, \tilde\mu] +$$
$$\frac{8}{3} m_3 \tilde\mu g_3^2 y_d^{pi2} \overline{a_d}^{pr} a_d^{i1r} \mathrm{LF}_{1,1,1,1,1,0}[m_3, m_{\tilde d}^{i2}, m_{\tilde d}^{r}, m_{\tilde q}^{i1}, m_{\tilde q}^{p}] +$$
$$\frac{8}{3} m_3 \tilde\mu g_3^2 y_u^{i1r} (a_d^{pi2} \overline{a_u}^{pr} - \tilde\mu^2 y_d^{pi2} \overline{y_u}^{pr}) \mathrm{LF}_{1,1,1,1,1,0}[m_3, m_{\tilde d}^{i2}, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, m_{\tilde u}^{r}] +$$
$$\tilde\mu y_d^{pi2} y_u^{i1t} \overline{a_d}^{pr} (-a_d^{sr} \overline{a_u}^{st} + \tilde\mu^2 y_d^{sr} \overline{y_u}^{st}) \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{t}, \tilde\mu] +$$
$$\tilde\mu^3 y_d^{si2} \overline{y_u}^{st} y_u^{pt} y_u^{i1r} \overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{r}, m_{\tilde u}^{t}, \tilde\mu]\Big)$$